

SUVAT Graphs, Distance in Time Intervals

A complete, evidence-led revision guide from Lampros Andrinopoulos's lesson of 18 July 2026

What this guide teaches

How to choose and control SUVAT equations; prove proportional graph shapes; find displacement within an accelerated time interval; decode the “second second” and “fourth second”; solve time-above-a-height problems; and keep signs, roots, units and rounding under control.

Specification alignment: Pearson Edexcel IAL Physics, Issue 3, WPH11/01, Mechanics statements 1–3.

Evidence status: lesson audio/visual evidence is controlled through 00:31:44. Material after 00:31:45 is used only for transcript-supported status statements; no later screen identity is claimed.

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1. Scope and evidence key

This guide separates what was observed in the lesson, what is controlled by an official paper or specification, and what has been independently recalculated. That matters because a correct revision guide must preserve both the learning process and the final physics.

LESSON EVIDENCE

The learner's attempt, Lampros's prompts and corrections, and visible board work through 00:31:44.

OFFICIAL CONTROL

Exact question identity, marks and answers where an official OCR or Pearson source was recovered.

INDEPENDENT CALCULATION

Arithmetic recomputed from the controlled values, using unrounded intermediate values.

SOURCE / MARKS UNKNOWN

Q04 and Q05 come from a tutor/"Kinematics" worksheet whose exact publication and marks could not be identified.

Boundary that must not be blurred. The recording is visually recoverable only through 00:31:44. Previous-lesson homework and projectile study-ahead are therefore listed later only as transcript-supported status. Their exact on-screen questions are not reconstructed.

Edexcel IAL specification map

Issue 3 requirement	Where this guide addresses it
Mechanics 1: use the four equations for uniformly accelerated one-dimensional motion	Every worked example and the equation-choice workflow
Mechanics 2: draw and interpret displacement-time, velocity-time and acceleration-time graphs	Q01 and the graph/interval diagrams
Mechanics 3: derive physical quantities from graph slopes and areas, including non-uniform cases	Graph interpretation summary and the endpoint-subtraction method

2. One-page exam summary

The four SUVAT equations

$$\begin{aligned}v &= u + at \\s &= \frac{1}{2}(u + v)t \\s &= ut + \frac{1}{2}at^2 \\v^2 &= u^2 + 2as\end{aligned}$$

Condition: acceleration must be constant over the modeled interval.

Symbols and SI units

Symbol	Meaning	Unit
s	displacement	m
u	initial velocity	m s^{-1}
v	final velocity	m s^{-1}
a	acceleration	m s^{-2}
t	elapsed time	s

If you remember nothing else:

1. Choose one positive direction and write signs before selecting an equation.
2. “Between two times” usually means $s(t_2) - s(t_1)$, not “put Δt into SUVAT”.
3. The *n*th second is the interval $n-1 \rightarrow n$.
4. Direct proportionality requires a straight line **through the origin**.
5. For a vertical projection with up positive, $a = -g$ throughout ascent and descent.
6. For “time above a height”, solve for both crossing times and subtract larger minus smaller—but only after checking that two relevant crossings exist.

Fast equation selection

Missing / unwanted variable	Best starting equation	Typical trigger
No displacement s	$v = u + at$	Time to maximum height
No acceleration a	$s = \frac{1}{2}(u+v)t$	Uniform acceleration with endpoint velocities
No final velocity v	$s = ut + \frac{1}{2}at^2$	Position at specified times
No time t	$v^2 = u^2 + 2as$	Maximum height or braking distance

3. SUVAT foundations and equation choice

3.1 The conditions come before the formula

SUVAT is not a collection of universal motion equations. It models **one-dimensional motion with constant acceleration**. For free fall near Earth's surface, use $g = 9.81 \text{ m s}^{-2}$ when the lesson or question does so and neglect air resistance unless told otherwise.

- **Displacement is signed.** It depends on your chosen positive direction.
- **Distance is scalar.** In a motion that reverses, total distance is not automatically the magnitude of one net displacement.
- **Time is an elapsed duration.** If you restart the clock at an interval boundary, then the interval's own initial velocity must also be reset correctly.

3.2 The five-line setup

- 1 Sketch the motion and mark the interval or endpoints.
- 2 Declare the positive direction: e.g. "up is positive".
- 3 Write s , u , v , a , t with units and signs.
- 4 Choose the equation that excludes the variable you do not know and do not need.
- 5 Substitute, solve, then check direction, magnitude, units and precision.

Textbook-backed habit: write down the known SUVAT quantities first, then select the equation. This prevents equation-hunting by appearance.

3.3 Graph connections

Graph	Gradient gives	Area gives
displacement-time	velocity	not a standard motion quantity
velocity-time	acceleration	displacement
acceleration-time	rate of change of acceleration ("jerk"), beyond this lesson	change in velocity

4. Q01 — proportional graph and free fall

OFFICIAL CONTROL

OCR A Level Physics A H556/01, June 2017, Q4, **1 mark**. Official answer: **A**. The recording begins during the explanation, so the very first setup is not preserved.

Prompt

A ball is dropped from rest. Air resistance is negligible. Its speed is v after falling distance h . Select the correct graph.

Lesson attempt: the learner chose A from the qualitative idea that greater fallen distance corresponds to greater speed.

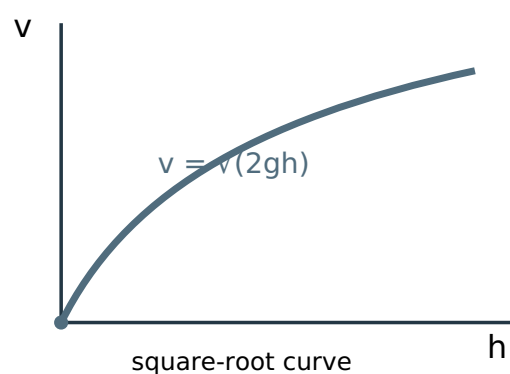
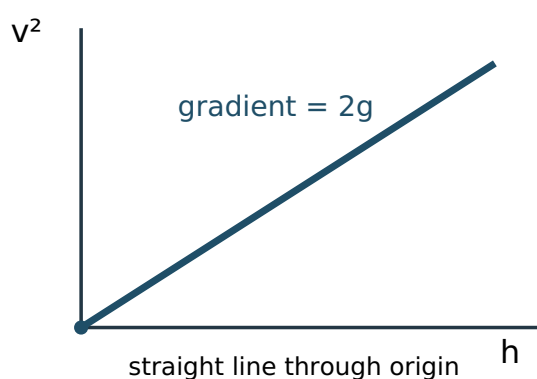
Lampros's correction / strengthening: "increasing" is not enough to prove the graph's exact form. A straight line alone also does not prove direct proportionality; it must pass through the origin.

Controlled derivation

Use the no-time equation because the graph connects speed and distance:

$$\begin{aligned}v^2 &= u^2 + 2as \\ u &= 0, a = g, s = h \\ \mathbf{v^2} &= \mathbf{2gh}\end{aligned}$$

Therefore $v^2 \propto h$. A graph of v^2 against h is a straight line through the origin with gradient $2g$. Equivalently, $v = \sqrt{2gh}$, so a graph of v against h is a square-root curve, not a straight line.



Do not confuse the straight-line graph of v^2 against h with the curved graph of v against h .

Exam trap: a line such as $y = kx + c$ with $c \neq 0$ is linear but not directly proportional. A negative-gradient line through the origin, e.g. $y = -kx$, can still be a proportional relationship.

5. Q02 — displacement in a time interval

OFFICIAL CONTROL

OCR H556/01, June 2017, Q12, 1 mark. Official answer: **B, 0.25 m.**

Prompt

An object is released from rest at $t = 0$. Air resistance is negligible. Find the distance traveled between $t = 0.20$ s and $t = 0.30$ s.

Lesson attempt: the learner first focused on $\Delta t = 0.10$ s and considered $v = \Delta s / \Delta t$.

Lampros's correction: velocity is changing, so a constant-velocity distance formula over 0.10 s is invalid. If you restart the clock at 0.20 s, you would need the non-zero velocity at that instant.

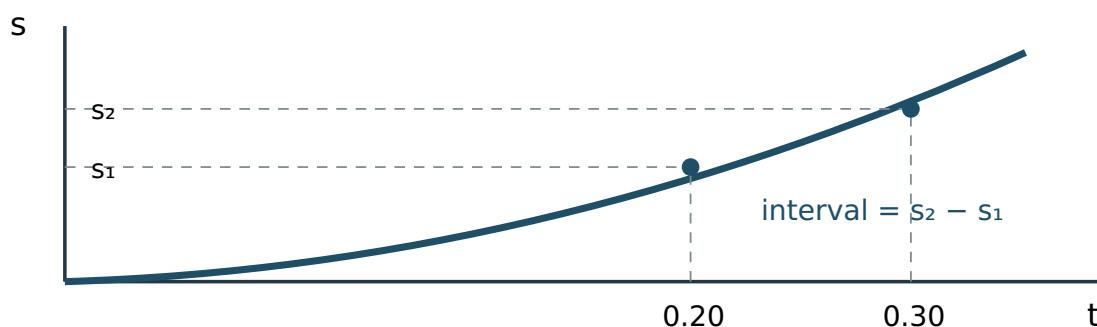
Reliable endpoint method

- 1 Use one clock from release and calculate the cumulative displacement at each absolute time.
- 2 With downward positive, $u = 0$, $a = +9.81 \text{ m s}^{-2}$, and $s(t) = \frac{1}{2}gt^2$.
- 3 Subtract the earlier cumulative displacement from the later one.

$$\begin{aligned}\Delta s &= s(0.30) - s(0.20) \\ &= \frac{1}{2}(9.81)[(0.30)^2 - (0.20)^2] \\ &= 0.2452500 \text{ m} \approx 0.25 \text{ m}\end{aligned}$$

INDEPENDENT CALCULATION

The full arithmetic above is reconstructed from the controlled values; the lesson emphasized the method more than every calculator line.



General interval formula from rest

$$\Delta s = \frac{1}{2}a(t_2^2 - t_1^2) = \frac{1}{2}a(t_2 - t_1)(t_2 + t_1)$$

This explains why equal-duration intervals get longer under positive acceleration: the factor $t_2 + t_1$ grows as time passes.

6. Q03 — “second second” and “fourth second”

OFFICIAL CONTROL

OCR Physics A H156/01 Sample Assessment Materials, Q9, 1 mark. Official answer:

A, 28 m.

Translation rule: the n th second means the one-second interval from $t = n-1$ to $t = n$. So the second second is $1 \rightarrow 2$ s; the fourth second is $3 \rightarrow 4$ s.

Prompt

A car accelerates uniformly from rest and travels 12 m during the second second. How far does it travel during the fourth second?

Lesson attempt: the learner was initially unsure whether “second second” meant $0 \rightarrow 2$ s. Later, after finding $s(4)=64$ m, that cumulative value was momentarily treated as the answer.

Lampros's correction: label the intervals on a timeline. The answer must be the change from 3 s to 4 s, not total displacement from 0 s to 4 s.

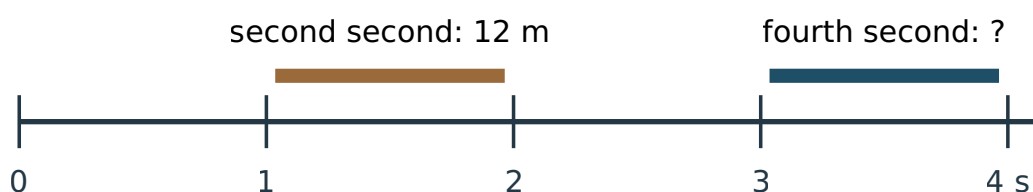
Step 1: use the second-second condition to find acceleration

Since the car starts from rest, $s(t) = \frac{1}{2}at^2$.

$$\begin{aligned}s(2) - s(1) &= 12 \\ \frac{1}{2}a(2^2) - \frac{1}{2}a(1^2) &= 12 \\ 2a - \frac{1}{2}a &= 12 \\ \mathbf{a} &= \mathbf{8 \text{ m s}^{-2}}\end{aligned}$$

Step 2: calculate the fourth-second displacement

$$\begin{aligned}s(4) &= \frac{1}{2}(8)(4^2) = 64 \text{ m} \\ s(3) &= \frac{1}{2}(8)(3^2) = 36 \text{ m} \\ \mathbf{s(4) - s(3)} &= \mathbf{28 \text{ m}}\end{aligned}$$



Reusable nth-second formula

Let each interval have duration $\tau = 1$ s. For constant acceleration, displacement during the n th interval is:

$$\begin{aligned}\Delta s_n &= s(n\tau) - s[(n-1)\tau] \\ &= u\tau + \frac{1}{2}a(2n-1)\tau^2 \\ &\quad \text{with } \tau = 1 \text{ s}\end{aligned}$$

When u and a are entered as SI numerical values and each interval is exactly one second, this is conventionally shortened to $s_n = u + \frac{1}{2}a(2n-1)$. For this car, $u=0$. The second-second distance is $\frac{1}{2}a(3)=12$, giving $a=8$; the fourth-second distance is $\frac{1}{2}(8)(7)=28$ m. Use the shortcut as a check, not as a substitute for understanding the interval or tracking units.

7. Q04 — time above a height

TUTOR / WORKSHEET SOURCE

Exact source unknown; **marks unknown**. The lesson presented a general derivation rather than an independently written learner solution.

General problem

A ball is projected vertically upward from $s=0$ with initial speed u . How long is it above a height h_0 ?

Choose upward as positive. Gravity remains downward for the entire flight:

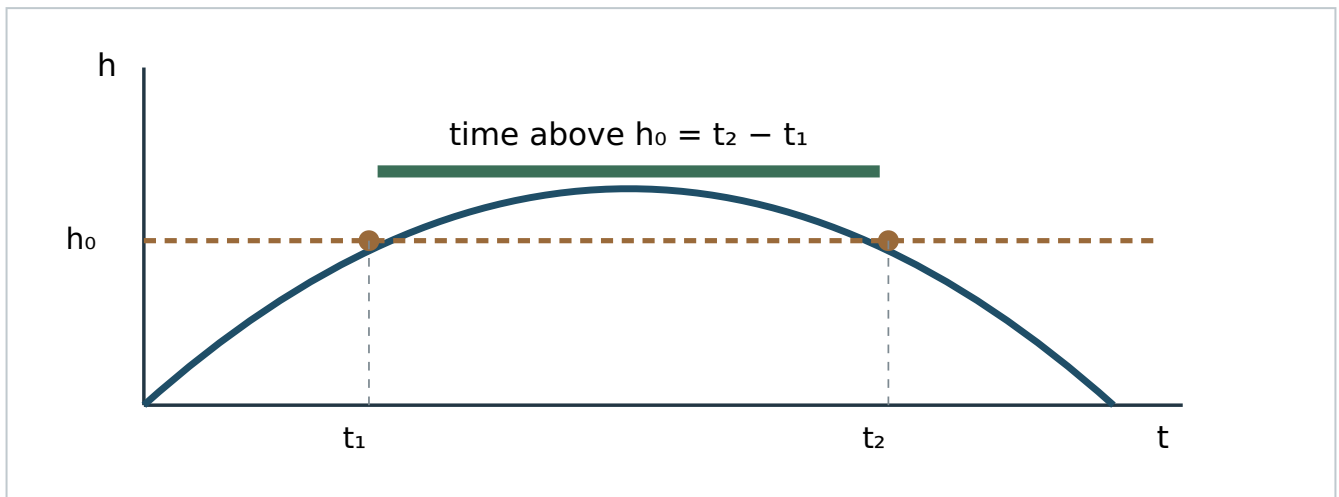
$$\begin{aligned}s &= ut + \frac{1}{2}at^2, \text{ with } a = -g \\ h_0 &= ut - \frac{1}{2}gt^2 \\ \frac{1}{2}gt^2 - ut + h_0 &= 0\end{aligned}$$

The two roots are the upward and downward crossings:

$$\begin{aligned}t_1 &= [u - \sqrt{(u^2 - 2gh_0)}]/g \\ t_2 &= [u + \sqrt{(u^2 - 2gh_0)}]/g \\ \text{time above } h_0 &= t_2 - t_1 = 2\sqrt{(u^2 - 2gh_0)}/g\end{aligned}$$

INDEPENDENT EXTENSION

The compact root-difference formula is derived from the lesson's quadratic method; it is not presented as a verbatim board result.



Conditions: when do two relevant roots exist?

- The maximum height is $h_{\max} = u^2/(2g)$.
- If $0 < h_0 < h_{\max}$, there are two positive crossing times after launch, so subtract them.
- If $h_0 = h_{\max}$, the two roots coincide and the ball is above that exact height for zero duration.
- If $h_0 > h_{\max}$, the discriminant is negative: the ball never reaches that height.
- If the requested height is below the launch level, one algebraic root may occur before $t=0$. Restrict the answer to the physical time domain stated in the question rather than subtracting roots blindly.

Do not split ascent and descent unless it helps. One equation with $a = -g$ already describes both. At the top, velocity is zero only instantaneously; acceleration is still $-g$.

8. Q05 — vertical projection at 10 m s^{-1}

TUTOR / "KINEMATICS" WORKSHEET

Exact source unknown; marks for Q05(a)–(c) are unknown. No official paper code, question number or mark allocation was recovered.

One sign convention for all parts: upward positive, $u = +10 \text{ m s}^{-1}$, $a = -9.81 \text{ m s}^{-2}$. Keep this convention through ascent and descent.

Q05(a): show the maximum height is about 5.1 m

Lesson attempt: the learner correctly identified $v=0$ at maximum height and selected the equation without time.

Lampros's correction: 5.1 m is a "show that" target, not a value to substitute. Use $g=9.81 \text{ m s}^{-2}$ in this Physics working.

$$\begin{aligned}v^2 &= u^2 + 2as \\0 &= 10^2 + 2(-9.81)s \\s &= 100/19.62 = 5.096839959... \text{ m} \approx 5.1 \text{ m}\end{aligned}$$

INDEPENDENT CALCULATION

Unrounded value shown before the requested "about 5.1 m" presentation.

Q05(b): find the time to maximum height

Lesson strategy: the learner deliberately chose a method independent of the part-(a) displacement, reducing error carry-forward. Lampros praised this.

$$\begin{aligned}v &= u + at \\0 &= 10 - 9.81t \\t &= 10/9.81 = 1.019367992... \text{ s} \approx 1.02 \text{ s}\end{aligned}$$

Depending on the precision convention demanded by a mark scheme, 1.0 s at 2 s.f. may also be the intended presentation. Do not round the intermediate before using it elsewhere.

Q05(c): find the time spent above 3 m

Lesson attempt: the learner formed the quadratic and then questioned root order and significant figures.

Lampros's correction: first check that 3 m is below the 5.1 m maximum, so two crossings exist. Then avoid the double-negative error: either write $s=ut+\frac{1}{2}at^2$ and substitute $a=-9.81$, or write $s=ut-\frac{1}{2}gt^2$; do not do both.

$$3 = 10t - 4.905t^2$$
$$4.905t^2 - 10t + 3 = 0$$

Provenance	Crossing times	Time above 3 m
TUTOR / BOARD ROUNDING	approximately 0.37 s and 1.68 s	about 1.3 s
INDEPENDENT FULL PRECISION	0.365540562... s and 1.673195421... s	1.307654859... s $\approx$ 1.31 s (or 1.3 s at 2 s.f.)

The labels t_1 and t_2 are arbitrary; the robust instruction is **larger root minus smaller root**.

Vertical-projection checks

- $t_{\text{top}} \approx 1.019$ s lies between the two 3 m crossing times, as it should.
- The two crossing times are symmetric about the top time: their average is approximately 1.019 s.
- The height equation has units of metres in every term: ut and $\frac{1}{2}at^2$ are both displacements.

9. Controlled homework — Pearson 8PH0/01 May 2016

Q13

OFFICIAL PEARSON CONTROL

Pearson Edexcel GCE AS Physics 8PH0/01, Tuesday 24 May 2016, Q13(a)–(b). Each subpart is **3 marks**. These are two of the three current-lesson items selected by Lampros.

Scope: the third selected current-lesson homework item is **unrecoverable**. It is not inferred or replaced here. The two subparts below are the only current-lesson homework questions whose identity, values, marks and official outcomes are controlled.

Q13(a) — braking distance scaling [3 marks]

Controlled task statement

A road train has braking distance 70 m at 15 m s^{-1} . Assuming the same braking force at all speeds, show that its braking distance at 25 m s^{-1} is about 190 m.

METHOD

The same braking force on the same vehicle mass gives the same magnitude of deceleration. During braking, use $v^2 = u^2 + 2as$ with final velocity $v=0$:

$$\begin{aligned} 0 &= u^2 + 2as \Rightarrow s = -u^2/(2a) \\ \text{For fixed deceleration magnitude, } s &\propto u^2. \\ s_2 &= 70(25/15)^2 = 194.444444\dots \text{ m} \\ \mathbf{s_2 \approx 194 \text{ m} \approx 190 \text{ m}} \end{aligned}$$

OFFICIAL RESULT

194 m; 194 m or 195 m accepted in the official control.

MARK-WINNING STRUCTURE

1. Relate braking distance to speed squared using constant deceleration.
2. Use the ratio $(25/15)^2$.
3. Obtain about 190 m with units.

Q13(b) — length of a road train [3 marks]

Controlled task statement

A car accelerates uniformly at 3.7 m s^{-2} while passing a stationary road train. Its initial speed is 30 m s^{-1} , and it reaches the 130 km h^{-1} speed limit as it passes the front of the cab. Calculate the road train's length.

METHOD

Convert the speed limit before using SUVAT:

$$\begin{aligned}v &= 130/3.6 = 36.111111... \text{ m s}^{-1} \\v^2 &= u^2 + 2as \\s &= [36.111111...^2 - 30^2]/[2(3.7)] \\s &= 54.596262... \text{ m} \approx 55 \text{ m}\end{aligned}$$

OFFICIAL RESULT 55 m; the controlled worked value is 54.6 m.

MARK-WINNING STRUCTURE

1. Convert km h^{-1} to m s^{-1} .
2. Select the no-time equation and rearrange for s .
3. Substitute unrounded speed, then round the final length to 55 m.

Other assigned / planned work — status only

Item	Reliable status	What is not claimed
Third current-lesson homework item	Selected, but unrecoverable	No identity, wording, values, marks or solution are invented
Previous Motion Graphs homework	Lampros selected prior-lesson items; the “challenge” was not required	Exact question identities are unavailable because this selection occurs after the visual-evidence boundary
Projectile motion	Deferred to the next lesson; optional study-ahead questions/resource were to be sent	No exercise wording or source is reconstructed

10. Formula, unit, sign and rounding traps

Trap 1: using Δt alone

If acceleration acts before the interval begins, the velocity at the interval start is not the original u . Prefer $s(t_2) - s(t_1)$ unless you correctly reset the full state.

Trap 2: cumulative vs interval

$s(4)$ is displacement from 0 to 4 s. The fourth-second displacement is $s(4) - s(3)$.

Trap 3: “straight line”

Linear does not mean directly proportional. Direct proportion requires zero intercept.

Trap 4: top of flight

At maximum height, $v=0$ but $a=-g$, not zero.

Trap 5: double negative

Use $s=ut+\frac{1}{2}at^2$ with $a=-g$, or $s=ut-\frac{1}{2}gt^2$. Never write a minus sign and then substitute negative g .

Trap 6: root order

Labels do not matter. Sort roots physically and calculate later minus earlier.

Trap 7: km h^{-1}

Divide by 3.6 to convert to m s^{-1} . Do this before squaring.

Trap 8: early rounding

Keep calculator precision through intermediate steps. Show the unrounded result, then the exam-appropriate rounded answer.

Dimensional checks

Expression	Unit check
ut	$(\text{m s}^{-1})(\text{s}) = \text{m}$
$\frac{1}{2}at^2$	$(\text{m s}^{-2})(\text{s}^2) = \text{m}$
$2as$	$(\text{m s}^{-2})(\text{s}) = \text{m s}^{-1}$, matching velocity squared
$\Delta v/a$	$(\text{m s}^{-1})/(\text{m s}^{-2}) = \text{s}$

11. Exam checklists and next actions

Before calculating

- ☐ Is acceleration constant over the modeled interval?
- ☐ Have I sketched the motion and marked the actual time interval?
- ☐ Have I declared a positive direction?
- ☐ Are all speeds in m s^{-1} , distances in m and times in s?
- ☐ Have I listed s, u, v, a, t with signs?

After calculating

- ☐ Did I answer interval displacement rather than cumulative displacement?
- ☐ If there are two roots, do they correspond to two physical crossings in the allowed time domain?
- ☐ Is the larger crossing time minus the smaller one positive?
- ☐ Does the direction/sign of the result make physical sense?
- ☐ Did I keep full precision until the final line, then include units and suitable rounding?

Next actions from this lesson

- 1 Complete Pearson 8PH0/01 May 2016 Q13(a) and Q13(b) without looking at the worked solutions; then compare line by line.
- 2 Make one timeline for each phrase: “during the second second”, “during the fourth second”, and “between 0.20 s and 0.30 s”.
- 3 Redo Q05(c) twice: first with the quadratic formula, then by using the symmetry of the height-time parabola as a check.
- 4 For every vertical problem, write “up positive $\Rightarrow a = -9.81 \text{ m s}^{-2}$ ” before substituting.
- 5 Keep the previous Motion Graphs homework and projectile study-ahead on the task list only by their known status; wait for the actual resources before attempting any exact question.

30-second verbal recall: “SUVAT needs constant acceleration. Intervals are endpoint differences. The nth second is n–1 to n. Up positive makes gravity negative. Time above a height is later root minus earlier root, after checking two crossings.”

12. Self-test

1. What extra feature turns a straight-line graph into evidence of direct proportionality?
2. A body starts from rest with constant acceleration 4.0 m s^{-2} . Find its displacement between 2.0 s and 3.0 s.
3. Translate “distance during the sixth second” into a time interval.
4. An object starts from rest and moves 7.5 m in its third second. Find its acceleration.
5. A ball is projected upward. What are its velocity and acceleration at maximum height if up is positive?
6. For a launch speed u , state the condition on h_0 for two positive crossing times of height h_0 .
7. Explain why inserting only 0.10 s into $s=ut+\frac{1}{2}at^2$ can fail for the interval 0.20 → 0.30 s.
8. Convert 108 km h^{-1} to m s^{-1} .

Answers

1. It must pass through the origin, so the relation has the form $y=kx$.

2. $\frac{1}{2}(4.0)(3.0^2-2.0^2)=10 \text{ m}$.

3. $t=5 \text{ s}$ to $t=6 \text{ s}$.

4. Third-second distance $=\frac{1}{2}a(2\times 3-1)=2.5a=7.5$, so $a=3.0 \text{ m s}^{-2}$.

5. $v=0$ instantaneously; $a=-g\approx-9.81 \text{ m s}^{-2}$.

6. For launch from $s=0$, require $0<h_0<u^2/(2g)$.

7. At 0.20 s the object already has non-zero velocity; restarting time without resetting u loses that state. Use endpoint positions or calculate the interval-start velocity.

8. $108/3.6=30 \text{ m s}^{-1}$.

13. Learner source notes

This guide was composed from a controlled lesson evidence set and the following reference hierarchy. These notes identify the authority without disclosing internal production material.

Source family	Use in this guide
Lampros Andrinopoulos lesson, 18 July 2026	Learner attempts, tutor corrections, teaching sequence, board-rounded Q05(c) values and task status. Audio/visual claims stop at 00:31:44; later status is transcript-only.
Pearson Edexcel International A Level Physics Specification, Issue 3	WPH11/01 Mechanics statements 1–3: uniformly accelerated equations; motion graphs; slope/area interpretation.
Edexcel Physics Year 1, Rectilinear Motion	SUVAT symbols and units, equation selection, free-fall sign convention, $v^2=2gh$, and the 10 m s^{-1} vertical projection reference example.
OCR H556/01 June 2017 Q4 and Q12, official question paper / mark scheme	Official identities, 1-mark controls and answers for Q01 and Q02.
OCR H156/01 Sample Assessment Materials Q9	Official identity, 1-mark control and answer for Q03.
Pearson Edexcel GCE AS Physics 8PH0/01, 24 May 2016, Q13(a)–(b)	Controlled homework values, 3 marks per part, and official outcomes 194/195 m and 55 m.
Independent recalculation	Q02 arithmetic; Q05 unrounded height, time and roots; Pearson Q13 full-precision calculations; dimensional and physical checks.

Declared uncertainties: Q04 and Q05 are tutor/worksheet material with source and marks unknown. The third selected current homework item is unrecoverable. Previous Motion Graphs homework and projectile study-ahead are status-only because the relevant part of the recording has no recoverable visual evidence.

Final formula sheet

Uniform acceleration

$$\begin{aligned}
 v &= u + at \\
 s &= \frac{1}{2}(u + v)t \\
 s &= ut + \frac{1}{2}at^2 \\
 v^2 &= u^2 + 2as
 \end{aligned}$$

Lesson transfer formulas

$$\begin{aligned}
 v^2 &= 2gh \\
 \Delta s &= s(t_2) - s(t_1) \\
 \Delta s_n &= u\tau + \frac{1}{2}a(2n-1)\tau^2, \tau = 1 \text{ s} \\
 h_{\max} &= u^2/(2g)
 \end{aligned}$$

Prepared as a learner-facing revision guide. No publication, upload, external message or revision-index action is part of this document.